Involutions and intersections in groups with determinant ±1

Human Compression project: N012 recovery

A component of the thesis recovery; distinct review pending

Let $q = p^f$ be odd and put

$$H_n(q) = \{g \in \mathrm{GL}_n(q) : \det g \in \{1, -1\}\}.$$

We recover the constructive and polytope assertions of Markovskaya's Theorems 4.2.1 and 4.2.2, and the part for dimensions at least five of the quotient assertions in Chapter 4 of [T]. The matrices and most root words are source constructions. Their generation arguments are replaced by the complete root-parameter theory [U]. A single five-dimensional calculation supplies the intersection property for every construction and every scalar quotient. The six-dimensional construction for $q > 3$ is a normalization of the already recovered journal matrices [L].

We use $t_{ij}(a) = I + aE_{ij}$, $x^y = yxy^{-1}$ and $[x, y] = xyx^{-1}y^{-1}$. Write $\tau e_i = e_{n+1-i}$ and $\mu e_i = e_{i+1}$, where $e_{n+1} = e_1$. For $\epsilon \in \{1, -1\}$, let C_ϵ reverse the first $n - 1$ coordinates and send e_n to ϵe_n. Then τC_ϵ is the monomial cycle with $e_i \mapsto e_{i+1}$ for $i < n$ and $e_n \mapsto \epsilon e_1$.

1 The common generation step

The following complete statements, with their proofs, are acquired from [U]. A monomial cycle and one nonidentity root on two cyclically consecutive coordinates reach every root subgroup: conjugate the seed around the cycle and apply $[t_{ij}(a), t_{jk}(b)] = t_{ik}(ab)$ along simple directed paths. If a subgroup $M \leq \mathrm{GL}_n(q)$, $n \geq 3$, meets every root subgroup, its root-generated subgroup is diagonally conjugate to $\mathrm{SL}_n(K)$ for a subfield $K \leq \mathbb{F}_q$. In particular it contains all determinant-one diagonal sign matrices. Opposite-root parameters a, b have product $ab \in K$. Finally a primitive $u \in \mathbb{F}_q^*$ satisfies $\mathbb{F}_p(u^2) = \mathbb{F}_q$. These are the particular forms needed below; none is inferred from testing finitely many fields.

Let $A = A(n, a)$ have off-diagonal entries $A_{21} = A_{n-1,n} = a \neq 0$ and no others. For $n = 5, 6$ its diagonal is negative at $1, n$ and positive elsewhere. For $n \geq 7$ it is negative at $2, 3, n - 2, n - 1$ and positive elsewhere. In all cases $A^2 = I$ and $A\tau = \tau A$. Put $D = \mathrm{diag}(1, -1, -1, 1, \ldots, 1)$. Whenever all roots have already been reached, $D \in M$ and direct multiplication gives

$$[D, A] = \begin{cases} t_{21}(2a), & n = 5, 6, \\ t_{21}(-2a), & n \geq 7. \end{cases} \tag{1}$$

This commutator isolates a root only after root recognition has supplied D; it is not used to justify the initial seed.

For $B = \tau$ or $-\tau$ and $E = BC_\epsilon$, transporting $t_{21}(b)$ first by B and then by E^2 gives $t_{12}(b)$. The two wraparound weights in the latter transport cancel. Consequently (1), with primitive $a = u$, forces $4u^2 \in K$ and hence $K = \mathbb{F}_q$. It follows that $\mathrm{SL}_n(q) \leq M$. If one generating matrix has determinant -1 and all have determinant in $\{1, -1\}$, then $M = H_n(q)$.

2 Five coordinates

For $q > 3$ choose a primitive $u \neq 2$. There are at least two primitive elements in the cyclic group of order $q - 1 \geq 4$, so this choice is possible. For $q = 3$ take $u = 1$; in that prime field the conclusion

about the parameter field is automatic. Use

$$A = A(5,u), \qquad B = -\tau, \qquad C = C_1, \qquad E = BC = -\mu.$$

These are involutions, A and B commute, and $\det B = -1$. The source words on pages 58–59 give a nonzero adjacent root as follows. Define

$$X = [A, A^E], \quad D_0 = [A, X^E], \quad Y = [X^E, D_0^{E^{-1}}], \quad T = Y^{E^3}, \quad J = [A^{E^2}, T].$$

Using $E_{ij}E_{ab} = \delta_{ja}E_{ib}$, their complete nonidentity entries are

$$\begin{aligned}
X &= t_{31}(-u^2)t_{41}(u^2),\\
D_0 &= t_{41}(u^4-u^3)t_{42}(u^3)t_{51}(-u^3)t_{52}(-2u^2),\\
D_0^{E^{-1}} &= t_{31}(u^3)t_{41}(-2u^2)t_{35}(u^4-u^3)t_{45}(-u^3),\\
Y &= t_{32}(-u^6+u^5)t_{42}(u^5),\\
T &= t_{15}(-u^6+u^5)t_{25}(u^5),\\
J &= t_{15}(u^6)t_{25}(-2u^5),\\
T^2J &= t_{15}\big(u^5(2-u)\big).
\end{aligned}$$

The factors in each displayed product commute. The last parameter is nonzero by the choice of u, including $u = 1$ in $\mathbb{F}_3$. The cycle E propagates this root to all roots. The common generation step gives $\langle A, B, C\rangle = H_5(q)$.

3 Seven coordinates and the uniform construction

For $n \geq 7$ take primitive $u \in \mathbb{F}_q^*$ and set

$$A = A(n,u), \quad B = \tau, \quad C = C_\epsilon, \qquad \epsilon = \begin{cases} 1, & n \equiv 0,3 \pmod 4,\\ -1, & n \equiv 1,2 \pmod 4. \end{cases}$$

Then $\det C = -1$, and A, B, C are involutions with $AB = BA$. Put $E = BC$ and

$$X = [A, A^E], \quad Y = (XA^{E^2})^2, \quad T = Y^E, \quad Z = [T, X], \quad R = [A, Z], \quad V = R^B, \quad W = [Z, V].$$

For every $n \geq 8$, direct multiplication gives

$$\begin{aligned}
X &= t_{31}(-u^2)t_{n-1,1}(\epsilon u^2),\\
Y &= t_{32}(-u^3)t_{41}(u^3)t_{42}(-u^4)t_{n-1,2}(\epsilon u^3),\\
T &= t_{43}(-u^3)t_{52}(u^3)t_{53}(-u^4)t_{n3}(\epsilon u^3),\\
Z &= t_{41}(u^5)t_{51}(u^6)t_{n1}(-\epsilon u^5),\\
R &= t_{n-1,1}(-\epsilon u^6), \qquad V = t_{2n}(-\epsilon u^6), \qquad W = t_{21}(-u^{11}).
\end{aligned} \tag{2}$$

Here is a uniform verification of the range of these identities. For the first two products use the eight distinct labels $1, 2, 3, 4, 5, n-2, n-1, n$. The diagonal negatives of A, A^E, A^{E^2} are respectively

$$\{2, 3, n-2, n-1\}, \quad \{3, 4, n-1, n\}, \quad \{4, 5, n, 1\}.$$

Their respective off-diagonal pairs are

$$(21 : u,\ n-1, n : u), \quad (32 : u,\ n, 1 : \epsilon u), \quad (43 : u,\ 12 : u).$$

All remaining coordinates are fixed. Multiplication on these labels gives X, Y. Conjugating the four entries of Y shifts each index one step without wraparound and gives T. The matrix-unit

rule then gives Z, R; reversal transports R to V, and a final commutator gives W. Both possible values of ϵ satisfy the same identities. The labels are distinct also when $n = 8$, so no unexamined small boundary is hidden in (2).

For $n = 7$ we have $\epsilon = 1$. The same definitions give the same X, T, Z, but the overlap $n-2 = 5$ changes the remaining products to

$$\begin{aligned} R &= t_{51}(-2u^6)t_{61}(-u^6), \\ V &= t_{27}(-u^6)t_{37}(-2u^6), \\ W &= t_{21}(-u^{11})t_{31}(-2u^{11}), \\ Q := [W, W^E] &= t_{31}(-u^{22})t_{41}(-2u^{22}), \\ J := [A, Q] &= t_{31}(2u^{22}), \qquad JQ^2 = t_{41}(-4u^{22}), \\ [(JQ^2)^{E^2}, V] &= t_{67}(8u^{28}). \end{aligned} \tag{3}$$

These are identities over $\mathbb{Z}[u]$, so the last parameter is nonzero in every odd characteristic. The square in JQ^2 is necessary: the printed cancellation on source page 67 omits it. We correct that individual expression, rather than infer a failure of the theorem. The adjacent root in (2) or (3), followed by the common generation step, proves $\langle A, B, C\rangle = H_n(q)$ for all $n \geq 7$.

4 Six coordinates

First let $q > 3$. We choose $v \in \mathbb{F}_q^*$ such that

$$-v \text{ is a square}, \qquad \mathbb{F}_p(v) = \mathbb{F}_q, \qquad h = v^2 + 4v \neq 0. \tag{4}$$

If $q \equiv 3 \pmod 4$, choose primitive $v \neq -4$. There are at least two primitive choices; both v and -1 are nonsquares, so $-v$ is square. If $q \equiv 1 \pmod 4$, choose $v = u^2 \neq -4$ with primitive u. The primitive-square result gives the field assertion. At $q = 5$ the choice is $v = 4$, whereas $-4 = 1$. For $q > 5$, the cyclic order $q - 1 \geq 8$ has at least four primitive elements, giving at least two distinct squares, so one avoids -4. In this case -1 is square as well. This proves (4).

Take $A = A(6, -1)$, $C = C_1$, and

$$B_0 = \operatorname{diag}(-v^{-1}, -v^{-1}, -v^{-1}, 1, 1, 1)\tau, \qquad B = \lambda B_0, \quad \lambda^2 = -v.$$

Then $B_0^2 = -v^{-1}I$, so $B^2 = I$; also $AB = BA$. Their determinants are $\det A = \det C = 1$ and $\det B = -1$. Conjugation by B equals that by B_0, and conjugation by $E = BC$ equals that by $E_0 = B_0C$.

The complete six-dimensional journal word from [L] is a Laurent-polynomial identity in v; its derivation does not require v primitive. In the present notation it reads

$$\begin{aligned} X = [A, A^E] &= t_{31}(-1)t_{51}(-v), \\ Q = [A^{E^4}, X] &= t_{51}(2v)t_{61}(-v^2), \\ [A, Q] &= t_{51}(-h), \\ (A^{E^4}t_{51}(h))^2 &= t_{61}(vh), \qquad t_{61}(vh)^E = t_{12}(vh). \end{aligned}$$

Every element on the left is in the generated group, since $t_{51}(h) = [A, Q]^{-1}$. By (4) the last root is nontrivial. Propagation gives every root. Now (1) gives $t_{21}(-2)$, and

$$\big(t_{21}(-2)^B\big)^{E^2} = t_{12}(2v).$$

Thus the parameter field contains $-4v$ and hence v; (4) makes it the full field. The determinant calculation finishes generation of $H_6(q)$.

For $q = 3$ use the original finite-field marking

$$A = A(6, 1), \qquad B = \tau, \qquad C = C_{-1}.$$

It suffices to give one adjacent elementary matrix. The following literal word in $a = A, b = B, c = C$, read left to right, has 131 letters and evaluates over $\mathbb{F}_3$ to $t_{21}(1)$; the line breaks have no algebraic meaning:

```
abcbcacbcacabcacacacbcbcacabcbcac
bacbcacacacabcacabcacbcacbcacbcac
babcacbcacabcacacacbcacbcacbcacbc
 acbcacbcacabcacacacbcacbcacbcacb
```

The exact word is also retained as an integer list. Its verification consists solely of multiplying the three displayed 6×6 matrices modulo three; the accompanying standard-library checker performs these 131 products and compares every entry with $I + E_{21}$. No group-order calculation or completeness of the search that found the word is a premise. Conjugating around BC and taking elementary commutators gives every root over the prime field. Hence the group contains $\mathrm{SL}_6(3)$; since $\det B = -1$, it is $H_6(3) = \mathrm{GL}_6(3)$.

5 One intersection calculation

The preceding constructions share a five-dimensional invariant subspace for A, C. Put $a_0 = A_{21} = A_{n-1,n} \neq 0$ and let $\sigma = -1$ for $n = 5, 6$, $\sigma = 1$ for $n \geq 7$. Write $\delta = -\sigma$ and $\epsilon = C_{nn}$. On the ordered basis

$$x = e_1, \quad y = e_2, \quad z = e_{n-2}, \quad w = e_{n-1}, \quad t = e_n,$$

the matrix $F = AC$ acts by

$$x \mapsto \delta w, \quad y \mapsto \delta z, \quad z \mapsto \delta y, \quad w \mapsto \sigma x + a_0 y, \quad t \mapsto \epsilon\sigma t + \epsilon a_0 w. \tag{5}$$

On the complementary coordinates it is a signed permutation whose permutation cycles have length at most two, so its fourth power is identity. Within (5), squaring gives

$$F^2 x = -x + \delta a_0 y, \qquad F^2 w = -w + \delta a_0 z,$$
$$F^2 y = y, \qquad F^2 z = z,$$
$$F^2 t = t + \sigma a_0 w + \epsilon\sigma a_0 x + \epsilon a_0^2 y.$$

Squaring again therefore proves, for every dimension in question,

$$F^4 = I + N, \qquad N = a_0^2(\epsilon E_{2n} - E_{n-2,n}), \quad N^2 = 0. \tag{6}$$

In particular $N \neq 0$. We next prove that a power F^j is monomial if and only if $4p$ divides j, in which case it is identity. For $j = 4l$, (6) gives $F^j = I + lN$. Its last column has three nonzero entries when $l \neq 0$ in $\mathbb{F}_p$, and it is identity otherwise. The other residues have two fixed nonzero entries that cannot be altered by right multiplication by $I + lN$, which changes only column n:

matrix	row	two entries (column:value)
F	2	$n-2 : \delta, \quad n-1 : a_0$
F^2	2	$1 : \delta a_0, \quad 2 : 1$
$F^{-1} = CA$	$n-2$	$1 : a_0, \quad 2 : \delta.$

The last row covers $F^{4l+3} = F^{-1}(I + (l+1)N)$. This proves the assertion for all integers j, as well as $|AC| = 4p$.

Since A, C are involutions, every element of $\langle A, C\rangle$ is F^j or F^jC. The matrix C is monomial, so the only monomial elements in that dihedral group are I, C. Both B and C are monomial in every construction. Hence

$$\langle A, C\rangle \cap \langle B, C\rangle = \langle C\rangle. \tag{7}$$

The three involutions are distinct: A is not monomial, while B, C have different permutation parts. With the order (A, C, B), their end involutions commute. The standard rank-three group–polytope correspondence applied to (7) therefore gives a regular polytope.

The order of BC is also explicit. For $n = 5$ it is $2n$ since $BC = -\mu$. For $n \geq 7$ its nth power is ϵI, and no smaller positive power has identity permutation part. Thus its order is n or $2n$. In the six-dimensional normalized construction, the product of the six monomial weights is one: its determinant is -1 and the permutation is a six-cycle of sign -1. Its order is therefore six. In the exceptional $q = 3$ construction, $(BC)^6 = -I$, so its order is twelve. In every case the type is $[4p, n]$ or $[4p, 2n]$.

Theorem 1. *For odd q and $n \geq 2$, $H_n(q)$ admits three generating involutions with two commuting if and only if $n \geq 5$. For every $n \geq 5$ it is the automorphism group of a regular polytope of rank three and type $[4p, n]$ or $[4p, 2n]$.*

Proof. The positive constructions and (7) prove every assertion for $n \geq 5$. For $n = 2, 3, 4$, the complete symmetric-square obstruction proved in the acquired central-quotient component [Z] excludes even the required generation. Its hypotheses are exactly odd q and the determinant-sign group. We retain that proof rather than duplicate it. □

Theorem 2. *Let $n \geq 5$ and let S be any subgroup of scalar matrices in $H_n(q)$. The images of the displayed marking give $H_n(q)/S$ a regular polytope of type $[4p, n]$ or $[4p, 2n]$. For the full scalar subgroup $S = Z(H_n(q))$, the type is $[4p, n]$.*

Proof. Suppose $x \in \langle A, C\rangle$ and $y \in \langle B, C\rangle$ have the same image. Then $x = ys$ for a scalar $s \in S$, so x is monomial and therefore $x \in \{I, C\}$. This proves the intersection property in the quotient. Moreover $\langle A, C\rangle \cap S = 1$, since C is not scalar. Its order $4p$ and the distinctness of the three involutions persist. The permutation part of BC is an n-cycle, so its image has order divisible by n. Its nth power is I or $-I$, hence the quotient order is n or $2n$; in the full scalar quotient it is exactly n. Generation passes to the quotient. □

6 Attribution and verification boundary

The recovered $H_n(q)$ classification and its polytope types are Markovskaya's results [T, pp. 56–69]. The matrices and seed words in dimensions five and at least seven are hers. Formula (3) repairs a missing square on the actual printed page 67. The six-dimensional normalized marking comes from [L], with a different parameter choice and scalar normalization; the prime-field word supplies the remaining case. The root theory is the acquired classical theory proved in [U]. The common five-dimensional argument expands the source's monomial-intersection calculation and makes its descent to every scalar quotient explicit. We make no literature-wide novelty claim for these arguments.

The exact polynomial checker verifies the displayed finite words, the uniform eight-label computation and the four sign choices in (5). The written matrix-unit proof, not a finite sample of dimensions, justifies the uniform ranges. The 131-letter prime-field identity has its separate complete multiplication certificate. Neither is a Lean formalization. This component does not settle the remaining small quotient cases or the full Chapter 3 routing audit, and gives no credit for the whole thesis, an open question or a measured compression factor.

References

[1] [T] I. A. Markovskaya, *Linear groups over finite fields and over the ring of integers as automorphism groups of regular polytopes of rank 3*, thesis, Siberian Federal University, 2025, 80-page supplied edition; supervisor Ya. N. Nuzhin.

[2] [L] *Three involutions in finite general linear groups*, reviewed recovery of I. A. Markovskaya and Ya. N. Nuzhin, *On generation of the groups* $\mathrm{GL}_n(q)$ *and* $\mathrm{PGL}_n(q)$ *by three involutions, two of which commute*, Trudy Inst. Mat. Mekh. UrO RAN 31(4) (2025), 247–259.

[3] [U] *Root parameters and fixed-space obstructions*, complete reviewed common paper accompanying [L].

[4] [Z] *Central quotients of the determinant-sign subgroup*, reviewed N012 component, including its complete low-dimensional generation obstruction and exact centre calculation.